

TONGRUI (JEREMY) QIAO

Saint John, NB

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Education

University of New Brunswick – Saint John

Bachelor of Science in Computer Science

• **Dean's List** – 2025/2026 (GPA: 3.8/4.3)

Expected Dec 2026

Saint John, NB

Technical Skills

Programming: Python, C, C++, Java, SQL, PHP

AI & Data: Reinforcement Learning, Agent-based Modeling, NLP, RAG, PyTorch, NumPy, Pandas, Vector Search

Web & Systems: HTML, CSS, JavaScript, MongoDB, Linux, Git

Experience

Software Engineer

Apr 2026 – Present

Inteplast Group

Saint John, NB

- Translated and localized industrial software interfaces, technical manuals, and pipeline documentation, ensuring accurate and consistent terminology across systems.
- Generated and managed QR codes for machinery documentation, linking physical equipment to digital instruction resources for streamlined operator access.
- Leveraged AI tooling to assist with automated code commenting and improving codebase readability.
- Collaborated with engineering teams to maintain alignment between localized content and evolving software specifications.

Undergraduate Tutor

Jul 2024 – Apr 2026

The Flora Beckett Math and Science Help Centre

Saint John, NB

- Tutored university students in mathematics, statistics, and computer science, explaining complex concepts and guiding problem-solving.
- Assisted students in troubleshooting programming errors and debugging code.

Undergraduate Teaching Assistant

Jan 2025 – Jan 2026

University of New Brunswick

Saint John, NB

- Graded assignments and supported first-year Computer Science students in introductory programming courses.
- Improved student understanding of core programming concepts.

Projects

Ontology-Guided Occupational Disease QA System — Python, RAG, NLP, LLM

Apr 2026

- Developed an ontology-guided Retrieval-Augmented Generation (RAG) system to answer occupational disease questions using the ILO Diagnostic and Exposure Criteria for Occupational Diseases published by the International Labour Organization, enabling evidence-based responses from a large medical reference document.
- Implemented a semantic retrieval pipeline including document chunking, embedding generation, and vector similarity search to retrieve relevant evidence before LLM generation, improving answer relevance and reducing unsupported responses.
- Built a structured evaluation dataset using expert-designed occupational disease questions provided by a medical specialist from Dalhousie University, and evaluated baseline LLM responses generated through Illinois Chat, a research LLM platform developed by the University of Illinois Urbana-Champaign.
- Designed a comparative evaluation framework to analyze baseline LLM responses, constrained prompting, and ontology-guided RAG approaches, enabling systematic comparison of model performance.
- Performed error analysis to identify hallucinated information, missing evidence, and incorrect ICD-10 medical classifications, helping reveal limitations of LLM-generated occupational health answers.

Intelligent Tic-Tac-Toe Agent — Python, AI Search, CNN, DNN

Jan 2026

- Developed an intelligent Tic-Tac-Toe agent using minimax search with alpha-beta pruning, reducing time complexity from $O(b^m)$ to $O(b^{m/2})$ and supporting scalable decision-making for $n \times n$ board sizes.
- Designed heuristic evaluation functions with depth-limited search to score non-terminal board states, incorporating pattern recognition for winning sequences, opponent threats, and positional advantages.
- Implemented a CNN(Convolutional neural network) as a learned evaluation function trained on self-play data to replace handcrafted heuristics, achieving over 86% validation accuracy across 20 training epochs.
- Built a DNN(Deep Neural Networks) policy network with an 18-dimensional binary input and softmax output to directly predict the next move without tree search, eliminating exponential search cost at the expense of optimality guarantees.

Movie Recommendation & Data Management System — MongoDB, JSON

Apr 2025

- Designed and structured JSON-based datasets using MongoDB document models to manage movie metadata and user rating information, enabling scalable storage and efficient data retrieval.
- Implemented analytical queries and aggregation pipelines on ratings and genre categories using NoSQL data modeling and indexing strategies, improving query efficiency and supporting recommendation filtering.

Licenses & Certifications

Generative AI Fundamentals — Databricks (Jun 2025) Microsoft AI & ML Engineering — Microsoft (Jun 2025) AI For Everyone — DeepLearning.AI (Aug 2025)